Journal of English for Academic Purposes 61 (2023) 101186

Contents lists available at ScienceDirect

Journal of English for Academic Purposes

journal homepage: www.elsevier.com/locate/jeap

Patterns and functions of *I* in academic writing: From a local grammar approach☆

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ARTICLE INFO

Keywords:
First person singular pronoun
Local grammar
Lexico-grammatical patterns
Semantic-functional patterns

ABSTRACT

This paper describes the co-selection of patterns and functions of the authorial *I* in soft science articles under the analytical framework of local grammar. Based on the taxonomy of Tang & John (1999), we generalized four functional categories in terms of the writer roles enacted by *I*: i.e. *I as architect*, *I as data analyser*, *I as recounter of research process*, and *I as opinion holder/originator*. We then identified the frequent semantic-functional patterns and lexico-grammatical patterns of *I* under each category of function. In so doing, we explored the connections among patterns and collocations of *I* and functions that *I* is used for. It is hoped that this paper would help novice soft science writers to gain a better understanding about how to use the authorial *I* more appropriately to construct authorial identity in academic writing.

1. Introduction

Traditional views hold that academic writing should adopt an impersonal writing style. As Smith (1996: 2–3) argued, "the first person should not appear in the scientific papers." Spencer and Arbon (1996: 26) also argued that "in the writing of traditional and formal style, *I* and *we* should not be used in the main body of the article." However, recent corpus-based studies have provided ample evidence on the common use of first person pronouns in academic writing of various genres (e.g. McGrath, 2016; Taylor & Goodall, 2019), disciplines (e.g. Isik-Tas, 2018; Yeo & Ting, 2014), and languages (e.g. Dueñas, 2007; Kim, 2009). As Hyland and Jiang (2017: 45) noted, "the convention of avoiding the first person to convey an impersonal stance, once a hallowed principle for style guide writers and science authors, now seems to be less rigidly adhered to." Compared with the first person plural *we*, which can be reader-inclusive (Harwood, 2005) and signify a reader-oriented hedge to mask the authorial presence, the first person singular *I* is totally writer-exclusive and makes the authorial presence explicitly visible. In the area of language and academic literacy education, teachers are sharply divided on whether *I* should be used in academic writing. As Taylor and Goodall (2019: 135) described, "some lecturers do not seem to mind it appearing in a piece of student academic writing, while others strongly object to it". Teachers' divergence makes the issue more problematic for student academic writing in the soft disciplines. However, the issue is nothing as simple as whether or not soft science writers should use *I*. It is necessary to look closely at the combinatory behaviour of *I* to better understand how soft science writers use *I* strategically and appropriately, as "native-like formulaicity builds upon knowledge about

☆ This research is supported by the Shanghai Planning Office of Philosophy and Social Science (No. 2021BYY001). We are grateful to Professor SU Hang, who kindly discussed the local grammar with us and made valuable suggestions. We also thank the editor and the anonymous reviewers for their constructive comments.

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https://doi.org/10.1016/j.jeap.2022.101186
Received 28 April 2022; Received in revised form 30 September 2022; Accepted 11 October 2022
Available online 19 November 2022
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specific phraseological patterns, regarding appropriate register, situational context, pragmatic functions and patterns of usage" (Wang & Zhang, 2021, p. 48).

The aim of a local grammar is to establish a connection between form and function in social contexts (Su, 2017). Compared with traditional or general grammars, which account for a function in terms of grammatical elements, local grammars offer a new approach for describing the realisations of a function in terms of both lexico-grammatical patterns and semantic-functional patterns. The semantic-functional pattern links at one end with the function by breaking the function into meaning elements, and at the other end with the lexico-grammatical pattern, where each meaning element maps onto a set of lexico-grammatical elements (see Section 2.2 for detailed description). By incorporating the semantic-functional pattern, local grammar pushes forward the description of the relationship between syntactic elements from the lexico-grammatical level to the discourse-semantic level (Su, 2019), and reveals more explicitly how form and function map onto and interact with each other, which in turn facilitates the establishment of a clearer connection between form and function. Following this line of thought, we have conducted the present study within the framework of local grammar to examine the actual use of the authorial *I* in soft science articles, with a view to characterizing the most prominent functions that *I* is used for and the most common local grammar patterns of *I* for each function. Accordingly, the research questions used to guide the analysis of this article are as follows:

(i) What are the main functions that the authorial *I* is used to realise in soft science article writing?
(ii) What are the local grammar patterns in terms of both semantic-functional patterns and lexico-grammatical patterns under each functional category of *I*, and what are the most frequent collocating verbs in each lexico-grammatical pattern?
(iii) How the patterns of *I* are co-selected with the functions of *I*?

2. Literature review

2.1. First person singular I in academic writing

A number of studies have been carried out to explore the discourse-pragmatic functions of first person singular *I* in various academic discourse. Functions of *I* in previous studies have been identified and classified from two main perspectives.

(1) Common research acts or rhetorical moves that reflect the development of a research paper (e.g. Dueñas, 2007; Harwood, 2005a; Hyland, 2002; Isik-Tas, 2018; Molino, 2010; Taylor & Goodall, 2019; Vassileva, 1998). To our knowledge, Vassileva (1998) was the first study to construct a taxonomy of ten functional categories for use of *I* in terms of the research acts that an *I*-sentence (the sentence where *I* occurs) in linguistics research articles may perform, but his/her taxonomy includes not only the authorial *I* but the unauthorial *I* as well. Hyland (2002) delineated the functions of authorial reference (e.g. the authorial *I*, *my*, *me*, *we*) and identified five functional categories of authorial reference in student reports and research articles, i.e. "Explaining a procedure", "Stating results or claim", "Elaborating an argument", "Stating a goal/purpose", and "Expressing self-benefits". Other functional taxonomies of the authorial *I* from this perspective are similar to Hyland (2002), but the specific functional identifications and classifications may vary slightly to better fit different research data. Harwood (2005a) refined Hyland's (2002) classification by adding two new functional categories for use of *I* based on the two corpora (a student corpus and an expert corpus) used in his study, and defined seven categories: "*I* to state results and claims", "*I* to elaborate an argument", "Methodological *I*", "*I* to state a purpose", "*I* to express self-benefits", "*I* to define terms", and "*I* as disciplinary servant". Molino (2010) set the threshold of "in at least one sub-corpus at a minimum frequency of one instance per 10,000 words" (ibid: 91) to determine which function of *I* (the function that an *I*-sentence achieves) can be considered "major", and Molino (2010) identified six major discourse functions for use of *I* in research articles: "Announcing goals or purposes", "Stating assumption", "Providing definitional clarifications", "Explaining procedures", "Making claims and elaborating an argument", and "Referring back to the text". Taylor and Goodall (2019) proposed a typology of eight functions for use of *I* based on the analysis of thirteen genres of student academic writing: "Acknowledging support and expressing gratitude", "Reflecting on self/actions", "Expressing intentions and decisions", "Stating a purpose", "Explaining a procedure", "Elaborating an argument", "Offering suggestions and recommendations" and "Stating results/claims". Isik-Tas (2018) took a step further and regrouped the functions of *I* in light of their chances of exposing the author to readers' direct challenges. Isik-Tas (2018) suggested that research acts such as "stating a goal or purpose" and "explaining a procedure" are low-stakes functions, while "stating personal opinions or knowledge claims" and "proposing a theory or approach" are high-stakes functions.

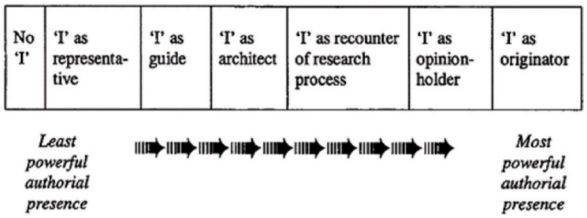

Fig. 1. Tang & John's functional taxonomy of first person pronouns (Tang & John, 1999, p. 29).

(2) Writer roles or writer identity in academic writing (e.g. Dong & Qiu, 2018; Leedham & Fernandez-Parra, 2017; McGrath, 2016; Raušová, 2018; Tang & John, 1999; Zareva, 2013). As noted by Tang and John (1999), "language does not serve merely as a tool to express a self that we already have, but serves as a resource for creating that self", the authorial *I* is found to be also a powerful means to signal writer roles and to construct writer identity in the text. Taking a cue from Ivanic's (1998: 307) idea of the continuum of *I*, Tang and John (1999) developed a taxonomy of six functions of first person in academic writing in terms of the roles that first person pronouns were mainly used to signal: "*I* as representative", "*I* as guide", "*I* as architect", "*I* as recounter of research process", "*I* as opinion holder" and "*I* as originator". The six functions were ordered along a continuum from the least powerful authorial presence to the most powerful authorial presence, as shown in Fig. 1. However, it should be noted that, although Tang and John (1999) labelled the six categories as "*I*", they actually considered all the various forms of the first person pronoun (i.e. *I, me, my, mine, we, us, our,* and *ours*) when building their taxonomy (ibid: 26).

Based on Tang and John's (1999) classification, other functional taxonomies of the authorial *I* have been developed in academic discourse across disciplines, text types, languages, writing expertise levels, and so on. Starfield and Ravelli (2006) proposed a taxonomy of five functions of *I* in Ph.D. theses in history and sociology: "*I* as guide or architect – to structure the discourse and state a purpose", "Methodological *I* – explains what was done (as researcher)", "*I* as opinion holder", "*I* as originator - claim maker", and "Reflexive *I*". The taxonomy added in Tang and John's (1999) classification the function "Reflexive *I*″, which is found to be commonly used by thesis writers to "create a powerful textual self that draws its authority through its explicit attention to autobiography and the use of *I* to locate the writer as subject within the research process" (Starfield & Ravelli, 2006, p. 231). Similarly, McGrath (2016) developed a taxonomy of six functions of self-mention in anthropology and history research articles: "Guide/architect", "Originator/claim maker", "Methodological *I*″", "Opinion holder", "Reflexive", and "Narrative". Zareva (2013) identified five categories of functions of the speaker-related *I* in TESOL graduate student presentations as "Guide", "Architect", "Recounter of research process", "Opinion holder", and "Originator". Leedham and Fernandez-Parra's (2017) devised five functional categories of *we* and *I* in Chinese, Greek and British students' assignments in engineering: "Guide", "Opinion holder", "Reflector", "Recounter", and "Representative".

To summarize, numerous studies have set out to explore the functions of the authorial *I* (e.g. Dueñas, 2007; Harwood, 2005a; Hyland, 2002; Isik-Tas, 2018; Leedham & Fernandez-Parra, 2017; McGrath, 2016; Molino, 2010; Raušová, 2018; Starfield & Ravelli, 2006; Tang & John, 1999; Taylor & Goodall, 2019; Vassileva, 1998; Yeo & Ting, 2014; Zareva, 2013). Although some studies have established links of the functions of the authorial *I* with the power of authorial presence signalled by the use of *I* (e.g. Hyland, 2002; Raušová, 2018; Tang & John, 1999), or discussed the rhetorical strategies underlying the use of *I*, such as the reasons why writers use *I* or the desired effects of such usage of *I* (e.g. Fortanet, 2004; Harwood, 2005b; Yeo & Ting, 2014), no study to date has attempted to account for the realisations of the functions of *I* in terms of semantic-functional patterns, lexico-grammatical patterns, and collocations from a local grammar approach.

2.2. Local grammar

Computational linguist Gross originated the concept of local grammar to capture shared features across instances of collocations and frozen sentences (Gross, 1993, p. 26). Later the concept of local grammar was developed by corpus linguists to deal with functions (e.g. Barnbrook & Sinclair, 2001; Hunston & Sinclair, 2000). As Hunston and Su (2017: 5) indicated, a local grammar is "a grammar of a discourse function" in specific registers and genres. Each local grammar "seeks to account for, not the whole of a language, but one meaning only" (Hunston, 2002, p. 178). Within the local grammar framework, a sentence of a particular function is analyzed and parsed into two kinds of pattern: the semantic-functional pattern, which is identified according to the semantic or functional role played by each constituent in the sentence, and the lexico-grammatical pattern, which is identified according to the words and word classes of each semantic or functional role. In each local grammar analysis, the semantic-functional pattern is matched with the lexico-grammatical pattern.

According to our knowledge, Hunston and Sinclair (2000) was the first to introduce the concept of local grammar to corpus linguistics and constructed the local grammar of evaluation. They first identified six lexico-grammatical patterns that are typically used to evaluate, and classified the sentences of evaluation into different categories of lexico-grammatical patterns accordingly. They then identified the semantic-functional pattern for each of the six lexico-grammatical patterns by using such meaning/function-specific labels as "evaluator", "evaluative category", "evaluating response", "thing evaluated" and "hinge". Table 1 shows some examples of the second lexico-grammatical pattern "*there* + link verb + *something/anything/nothing* + adjective group +*about/in* + noun group/-ing clause" in Hunston and Sinclair's (2000) local grammar analysis of evaluation.

Table 1
Local grammar analysis of the second lexico-grammatical pattern of evaluation (Hunston & Sinclair, 2000, p. 86).

there	link verb	*Hinge*	*Evaluative category*	*Hinge*	*Thing evaluated*
		something/anything/nothing	adjective group	*about* or *in*	noun group or -ing clause
There	's	something	rather appealing	about	being able to spend the evening in a town.
There	is	nothing	sacrosanct	about	this unit of analysis.
There	is	something	ironic	in	seeing the Dalai Lama
There	isn't	anything	romantic	about	trying to do a love scene …
There	is	something	very American	about	the National Archives collection …

Hunston and Sinclair's (2000) investigation provided important methodological and theoretical guidance for later research on how to apply the local grammar approach to dealing with functions. Based on the data of 44 adjective complementation patterns, Hunston and Su (2017) derived the local grammar of evaluation consisting of 22 "Analyses" that were grouped into five broad categories. In a reversal from Hunston and Sinclair (2000), Hunston and Su (2017) made the local grammar analysis of evaluation by using semantic-functional patterns as the first-level classifier and summarizing lexico-grammatical patterns for each of the semantic-functional patterns. Table 2 shows some examples of "Analysis 1" (*Hinge* + *Evaluation* + *Target*) in Hunston and Su (2017).

Whichever of the above approaches to analysis is adopted, the local grammar always accounts for a function by first establishing the mappings between lexico-grammatical patterns (such as "*it* v-link ADJ that-clause", "*it* v-link ADJ wh-clause", "*it* v-link ADJ to-inf." etc. in Table 2) and semantic-functional patterns (such as "Hinge + Evaluation + Target" in Table 2) to constitute meaning-form pairings, and then matching a set of such meaning-form pairings with the function. By so doing, local grammars describe the relationship between syntactic elements not only in lexico-grammatical categories but also in semantic-functional categories. In this sense, local grammars are more informative for language description than traditional or general grammars, as local grammars identify a syntactic element in terms of both its semantic-functional and lexico-grammatical categories, rather than simply (lexico-)grammatical category. This is more fully addressed in Su and Zhang (2019).

> What is noteworthy about local grammars is that they push the emphasis on phraseological/lexical-grammatical patterning towards an emphasis on discourse-semantic patterning, which further brings our understanding of text organisation and development to a new level. To be specific, traditionally, researchers and education practitioners consider text organisation and development in terms of patternings of lexical-grammatical forms; in contrast, research on local grammars allows us to understand text organisation and development from a relatively new perspective and at a higher level of abstraction: what we see in text is not just the strings of co-occurring phraseological forms but also strings of discourse-semantic patterns.
>
> (Su & Zhang, 2019, p. 15)

Two approaches can be distinguished among previous studies, regarding how to apply local grammar analyses to linguistic descriptions; one starts from a particular function and the other from a specific form.

Many studies on local grammars adopted the first approach (hereafter the "function-driven approach"), that is, taking one particular semantic or functional phenomenon as the starting point (e.g. Allen, 2005; Barnbrook, 2002; Bednarek, 2008; Cheng & Ching, 2018; Hunston & Sinclair, 2000; Hunston & Su, 2017; Su, 2017; Su & Zhang, 2019). Both Hunston and Sinclair (2000) and Hunston and Su (2017), as mentioned above, were studies of this type. Apart from evaluation, a local grammar approach is replicable for other functions, and in a similar vein, a number of studies developed the local grammars of different functions, for example, a local grammar of definition (Barnbrook, 2002), a local grammar of cause and effect (Allen, 2005), a local grammar of affect (Bednarek, 2008), a local grammar of request (Su, 2017), a local grammar of disclaimer (Cheng & Ching, 2018), and a local grammar of apology (Su & Wei, 2018). Those studies focused on the description of one meaning or function only and have demonstrated the validity and applicability of a local grammar approach to describing different functions. Studies of this type have suggested that local grammars have a noticeable advantage over general grammars in describing specific functional domains, as they are more systematic and detailed, and more straightforward (Su, 2017, p. 74).

The second approach to developing local grammars takes one particular lexical or grammatical form as its starting point (hereafter the "form-driven approach"). This approach is in line with Gross's (1993) motivation to originate the concept of local grammar to analyze restricted linguistic forms. The form-driven approach was developed as a complement to the function-driven approach to suit different research needs and for more thorough local grammar descriptions. As pointed out by Su (2018: 3), the form-driven approach can facilitate the development of local grammars "in a more systematic and principled manner". To the best of our knowledge, four studies to date have made local grammar analyses by using the form-driven approach. Sinclair (2007: 38–46) examined the local grammars of the multi-word lexical item with *sever* as its invariable core and made the case that "local grammars can better account for multi-word lexical items which exhibit variation as is often the case" (Warren & Leung, 2016, p. 3). Based on Sinclair (2007), Warren and Leung (2016) extended the theory and application of local grammars to identify the patterns of co-selection found in the four collocational frameworks *any* … *may*, *may* … *any*, *any* … *shall*, and *shall* … *any*. Su (2018) and Su (2019) demonstrated the possibility and feasibility of using lexico-grammar patterns as the input to develop local grammars; the former developed local grammars of the pattern "V n *on* n" for eight functions, and the latter developed the local grammars of the pattern "V n *as* n" for three broad meaning

Table 2
Local grammar analysis of the "Analysis 1" of evaluation (Hunston & Su, 2017, p. 17).

Analysis 1: Hinge + Evaluation + Target			
Element	Hinge	Evaluation	Traget
Pattern			
***it* v-link ADJ that**	It is	awful	that it should end like this
***it* v-link ADJ wh-**	It's	understandable	why they hate the sight of him
***it* v-link ADJ to-inf**	It would be	selfish	to marry
***it* v-link ADJ ing**	It was	ridiculous	putting him behind bars
it* v-link ADJ *about	It's	too bad	about the reviews
there* v-link sth ADJ *about	There's	something strange	about Mr Ross

groups. One of the main conclusions in both studies is that lexico-grammatical patterns are "a good starting point for developing local grammars, because searching lexico-grammatical patterns facilitates the identification of specific meanings in corpora and local grammars are dedicated to describing such meanings." (Su, 2019, p. 3).

While there have been a small number of studies to illustrate the feasibility of developing local grammars of restricted linguistic forms, there has been no attempt so far to make local grammar analyses of the function-rich word, the authorial *I*, and look into the co-selection features or combinatory behaviour of the word when it fulfils different functions. In view of this, based on a careful examination of a two million word corpus of soft science articles, our study first identifies the four main functions that an *I*-sentence may realise, and then examines the most frequent lexico-grammatical patterns and semantic-functional patterns for each function of the authorial *I* from a local grammar approach, with a view to characterizing the co-selection potential of functions and patterns of *I* in soft science articles.

3. Research methodology

3.1. The corpus

We based our study on the CARE corpus (Wei & Zhang, 2020). CARE (Collection of Academic Research Essays) is a discipline-specific comparable corpus of research articles published from the years 2003–2013, written either by English native speakers (CARE-ENS) or by Chinese native speakers (CARE-CNS). Please refer to Wei and Zhang (2020) for a more detailed description of CARE. CARE-ENS contains 2640 research articles written by English native speakers. It covers eight soft science disciplines, with reference to the notion of soft science identified by Kolb (1981: 241); they are economics and management, education, history, law, linguistics, political science, psychology, and sociology. The overall corpus size of CARE-ENS is eight million tokens, and the first person singular *I* occurs 8872 times throughout the corpus. As will be described in Section 3.3, the identification of the references and functions of *I* is a hugely labour-intensive process. The amount of 8872 instances was considered to be too big to handle manually. Hence, for the convenience of operation, we generated eight smaller groups of texts, each of which have roughly the same size of 250, 000 tokens, from the eight soft disciplines of CARE-ENS, respectively, and formed a small corpus, CARE-Sample, which was used to examine the functions and patterns of *I* in soft science articles. CARE-Sample contains 177 research articles with a manageable size of 2, 001,412 tokens and 34,363 types, and its standardised type-token ratio amounts to 26.55%.

3.2. Analytical framework

The framework applied to the analysis of the authorial *I* is two-layer: (1) the identification of functions with a view to the writer role enacted by *I* in discourse, and (2) the determination of the local grammar for each function in terms of semantic-functional patterns, lexico-grammatical patterns, and collocations.

At the first layer, our functional categorization of the authorial *I* draws on Tang and John's (1999) taxonomy. However, as mentioned before, Tang and John (1999: 26) built their taxonomy by considering a wider set of first person pronouns and therefore included in their taxonomy the uses which relate to items such as *we*, *us*, *our*, etc., rather than to *I*. With this in mind, we scrutinized the six functions categorized in their taxonomy, i.e. "representative", "guide through the essay", "architect", "recounter of research process", "opinion holder" and "originator", and filtered for the functions which can be realised by *I*. Specifically, we made the following modifications.

(1) According to Tang and John (1999), the role of "representative" is the usage of "a generic first person pronoun, usually realised as the plural *we* or *us*, that writers use as a proxy for a larger group of people" (ibid: 27). This writer role signals collective ownership and is most unlikely to be undertaken by the authorial *I* in research articles, therefore, we removed the function from our classification.

(2) The role of "guide through the essay" is "always implicitly or explicitly accompanied by the reader, this particular usage of the first person pronoun is usually realised in the plural, inclusive form of *we* or *us*" (Tang & John, 1999, p. 29). In this sense, the role refers to both authors and readers and unlikely involves the authorial *I*. However, Tang and John (ibid: 29) also indicate that "this particular role is often signalled explicitly by the use of verbs like *see*, *note*, and *observe* (in Halliday's (1994) terms, mental processes of perception, specifically visual perception)". We hand-checked the pairing of the authorial *I* with the verb of visual perception in our data and found dozens of such cases, in most of which the authorial *I* is not used to guide readers through the text, but to analyze data or result, such as in the example sentence *However, I* ***observe*** *neither private benefits nor managerial expertise*. Therefore, we removed the role of "guide through the essay" from our taxonomy but added a new role of *I* into it, i.e. "data analyser", which foregrounds the person who gleans useful information or result (factual) through the observation and analysis of data. Unlike the roles of "opinion holder" and "originator", which are both associated with expressing a view, the role of "data analyser" emphasizes on extracting factual information from data. As in the above example, the result of the observation *neither private benefits nor managerial expertise* is not an opinion but a fact.

(3) Tang and John (1999: 34) suggested that the role of "opinion-holder" and the role of "originator" would better be discussed together, because "there is often a natural progression from the role of opinion-holder (expressing an opinion of someone else's views) to the role of originator (venturing one's own take on an issue) in mature academic writing". In line with their view, we merged the two roles into one, which we tentatively call "*I* as opinion holder/originator".

Based on the above modifications to Tang and John's (1999) taxonomy, we finally constructed a taxonomy of four functions for the authorial *I* in soft science articles, i.e. *I as architect*, *I as data analyser*, *I as recounter of research process*, and *I as opinion holder/originator*.

At the second layer, no local grammar of self-mention, realised by the authorial *I* in particular, has yet been proposed based on our knowledge. We therefore developed a framework for local grammar analyses of the authorial *I*. As noted earlier, a local grammar analysis involves the identification of both semantic-functional patterns and lexico-grammatical patterns. The identification of two types of pattern is highly consistent cross local grammars of different functions, except that the categoric labels for meaning elements in semantic-functional patterns may vary depending on the specific function under study. For example, Hunston and Su (2017: 15) labelled the meaning elements in their local grammar analyses of evaluation as "evaluator", "evaluation", "evaluative act", "target", "hinge", "action", and "specifier". Su (2017) labelled the meaning elements in his analyses of request as "requester", "requestee", "requested action", "intended action", "hinge", "indicator", and "mitigator". Following the approach and labelling convention of local grammar analyses, we tentatively proposed the terms for meaning elements in semantic-functional patterns for each function of *I*. The terms are illustrated in Table 3 below. One term that needs to be explained here is "hinge". The concept of "hinge" varies across the

Table 3
Functional/meaning elements for the local grammar of each function of *I*.

Role-Specific Local Grammar Terms:	
Function 1: *I as architect*	
architect :	the one who architects (compose or write) the article (like an architect who constructs the building). In our study, the architect is always the first person singular *I*.
architecting action :	the action that the architect (writer) performs, intends to perform, or has performed. For example, *In the fifth section, I will* ***review*** *the close similarity between the statistical physics of ideal gases.*
thing architected :	the content of the article constructed by the author. For example, *In the fifth section, I will review* ***the close similarity*** *between the statistical physics of ideal gases.*
Function 2: *I as data analyser*	
data analyser :	the one who gleans useful information or result (factual) through the observation and analysis of data. In our study, the research processor is always *I*.
analysing action :	the action that the data analyser performs to obtain factual results. For example, *I* ***observe*** *neither private benefits nor managerial expertise.*
thing found :	the factual information or results found out based on analysis of data, which can then be used as a basis for agreeing, disagreeing, or putting forward a claim, view, etc. For example, *I observe* ***neither private benefits nor managerial expertise****.*
Function 3: *I as recounter of research process*	
research processor :	the one who advances the research process. In our study, the research processor is always *I*.
processing action :	the action that the research processor performs, intends to perform, or has performed. For example, *I* ***use*** *the following multinomial logistic model to assess the quantity of political information.*
thing processed :	objects processed by the research processor or procedures completed by the research processor. For example, *I use* ***the following multinomial logistic model*** *to assess the quantity of political information.*
Function 4: *I as the opinion-holder/originator*	
opinion-holder/originator :	the one who either shares an opinion, view, attitude, etc. or presents a new idea or knowledge claim. In our study, the opinion-holder/originator is always *I*.
opinion-holding/originating action :	the action that the opinion-holder/originator performs. For example, *However, I would* ***argue*** *that what is currently needed is a critical theory of place naming …*
opinion	either the view, attitude or idea shared by the opinion-holder or the new idea or knowledge claim put forward by the originator. For example, *However, I would argue that* ***what is currently needed is a critical theory of place naming*** *…*
Common Local Grammar Terms Across Roles:	
hinge :	the element that links different functional elements. Negation, mitigator and initiative (in italics) could be said to be finer-grained identifications of the hinge element featured in our local grammar analyses.
negation:	the element that negates, or reverses, the meaning of a statement or action. For example, *I would* ***not*** *consider the exercise so called 'sentence combining' as a communicative one ever.*
mitigator:	the element that mitigates the face-threatening act of expressing a view, opinion, etc. For example, *However, I* ***would*** *argue that what is currently needed is a critical theory of place naming …*
initiative:	the element that indicates how committed the author is to their research decisions or research choices. For example, *I* ***chose to*** *take a step back and examine one of the untested assertions underlying the field.*
specifier :	the element that specifies details of the performance of an action. The elements in italics could be said to be finer-grained sub-divisions of the specifier element.
aim :	the element indicating the purpose of the performance of an action or the aim to be realised. For example, *In the next section, I explore the potential of the concept of habitus* ***to explain the difficulty changing folk pedagogies****.*
comparator:	part of a statement of similarity or difference For example, *I match subsidiaries with debt outstanding* ***to divisions of firms****, using business segments in the Compustat database to proxy for divisions.*
manner:	the element relating to by which means the "thing" is generated. For example, *I estimate them* ***by subtracting capital expenditures from end-of-period assets and adding depreciation****.*
locator:	the element that indicates where or when the author performs, will perform or has performed an action. For example, ***In the fifth section,*** *I will review the close similarity between the statistical physics of ideal gases.*
scope:	the element that sets a restriction on the scope of the performance of an action. For example: *Then,* ***for each industry in each year****, I take the average of these standard deviations, weighting by each firm's beginning-of-period total assets.*
topic:	a specific entity or domain that the author talks about or shares an opinion on. For example, ***The point*** *I want to make here is that in view of the problems facing the Hughes …*

local grammar studies. Some studies defined the term as a grammatical linking operator (e.g. Barnbrook & Sinclair, 2001; Hunston & Sinclair, 2000), but others have seen the term as a functional linking operator (e.g. Cheng & Ching, 2018; Hunston & Su, 2017). This article takes the latter position and uses it to label the element that links different meaning elements in a semantic-functional pattern. In this terminology, "hinge" consists of not only grammatical words but also content words. For example, in the sentence ***I chose to examine*** *one of the untested assertions underlying the field*, which is used to describe the research process, *I* functions as the "research processor", *examine* is the "processing action", and *chose to* is the "hinge" that links the "research processor" with the "processing action".

3.3. Procedures

Data processing and analysis followed the three steps below.

First, we POS tagged all the texts in the CARE-Sample corpus with the tagging software CLAWS and its C7 tagging set, to distinguish between the first person singular *I* and the signs or symbols that have the same form with "I" (such as the Roman numeral "one"). The first person singular *I* was tagged as "I_PPIS1" by the software. Then we used Wordsmith Tools 6 to retrieve all the concordances which contain the tag "I_PPIS1" from the corpus, and obtained a preliminary output of 1821 concordances of the first person singular *I*.

Second, we noticed that in many cases *I* occurs in the citations, such as quoting the original words of subjects in experiments. *I* of this type is not author-related, so we read carefully the context where each instance of the first person *I* occurs, and determined whether *I* in each concordance refers to the author or not. After performing this step, we extracted 696 concordances of the authorial *I* from the corpus for further analysis.

Third, we applied the two-layer framework, as described in 3.2, to the analysis of the functions and patterns of the authorial *I* in soft science articles. To be more specific, we performed the following sub-steps. (1) We identified the function of each *I*-sentence by referring to the local textual environment or even to the extended context in which *I* occurs, and classified the 696 instances of the authorial *I* into four functional categories: *I as architect of the essay*, *I as data analyser*, *I as recounter of research process*, and *I as opinion holder/originator*. (2) We identified the semantic-functional pattern of each *I*-sentence under each functional category above, by using the labels of meaning elements as shown in Table 3. (3) We generalized the frequent lexico-grammatical patterns for each semantic-functional pattern and the most common verbs collocating with *I* in each lexico-grammatical pattern. We categorized the collocating verbs of *I* into three groups by essentially following Hyland and Tse's (2005: 130) identification of the three kinds of activity that verbs can refer to, i.e. research acts, which represent experimental activities or actions carried out in the real world (e.g. *demonstrates, shows*); cognition acts, which are concerned with the researcher's mental processes (e.g. *assume, believe*); and discourse acts, which focus on the expression of cognitive or research activities (e.g. *report, conclude*).

4. Results and analysis

4.1. Local grammar analyses of the four functions of I

Tables 4–7 reveal the differences in the frequency distribution of functions of the authorial *I*. It is shown that *I as recounter of research process* is the biggest functional type, with 302 instances accounting for 43.39% of the 696 occurrences of *I* in the corpus. This identity role is prevalent in the author's description of the various steps undertaken in research or the strategies and techniques utilized in the data collection and processing. Such an extensive use of *I* reveals that, compared with other identity roles of *I*, the soft science writers preferentially use *I* to manifest their authorial identity as designer or conductor of research and decision-makers in the research process. As Zareva (2013: 79) argued, "they probably saw their self-mention in reference to the details of their research inquiries as illuminating the objectivity and depth of their thinking process and, consequently, validating their opinions, beliefs, conclusions, and findings drawn from their research". The role of *I as opinion-holder/originator* ranks second, with 227 instances (32.62%). The soft science writers front this role with the authorial *I* to share opinions, to make claims, or to originate new ideas. According to Tang and

Table 4
Local grammar analyses of I as architect of the essay.

Function 1: *I as architect* (131 instances)	**(1).** Semantic-functional pattern:	**locator**	**architect**	**(hinge)**	**architecting action**	**thing architected + (aim)**	
	Lexico-grammatical pattern: (96 instances)	text deixis/ time reference/ listing adverbial	*I*	(modal verb/ verb of intention)	research/discourse act verb (common) cognition act verb	nominal + (to-inf) / wh-clause	
	Examples:	*In the next two sections,*	*I*	*will*	*describe*	*how the single- and dual-process assumptions fair when implemented in process-level models.*	
		Next,	*I*		*consider*	*the direction of partisan change, to distinguish whether one changes partisanship…*	
	(2). Semantic-functional pattern:	**hinge**	**architect**	**(hinge)**	**architecting action**	**locator**	**thing architected**
	Lexico-grammatical pattern: (17 instances)	*as*	*I*	(modal verb)	research/discourse act verb	text deixis / time reference	main clause
	Examples:	*As*	*I*		*discuss in more details*	*in the next section,*	*the last two results imply that…*
		As	*I*	*have*	*noted*	*above,*	*language challenges marginalised them in lectures,…*
	(3). Semantic-functional pattern:	**thing architected**		**(hinge)**	**architect**	**architecting action**	**locator**
	Lexico-grammatical pattern: (16 instances)	nominal		(that)	*I*	discourse act verb	text deixis
	Examples:	*The results*			*I*	*present*	*here*
		The results of the models		*that*	*I*	*present*	*above*

Table 5
Local grammar analyses of I as data analyser.

Function 2: ***I as data analyser*** **(36 instances)**	**(1).** Semantic-functional pattern:	**specifier**	**data-analyser**	**(hinge - negation)**	**analysing action**	**thing found + (specifier)**
	Lexico-grammatical pattern: (35 instances)	linking adverbial	*I*	(negative verb)	verb of perception/ verb of verification	object clause/ noun + (appositive clause)
	Examples:	*However,* (contrast)	*I*		*observe*	*neither private benefits nor managerial expertise.*
		In untabulated results, (scope)	*I*		*confirm*	*that when I use segment fixed effects, all my coefficients have the predicted signs but also higher standard errors.*
		Hence, (result or inference)	*I*	*fail to*	*find*	*evidence that firms use subsidiary debt to trade off between optimizing individual segment capital structures and coinsurance.*

Table 6
Local grammar analyses of I as recounter of research process.

Function 3: ***I as recounter of research process*** **(302 instances)**	**(1).** Semantic-functional pattern:	**(specifier)**	**research processor**	**(hinge - initiative)**	**processing action**	**thing processed + (specifier)**
	Lexico-grammatical pattern: (234 instances)	(linking adverbial)	*I*	(verb of decision/ intention)	research act verb	nominal + (aim/method/scope/comparator/etc.)
	Examples:	*therefore* (result/inference)	*I*	*decided to*	*utilize*	*the University's mass-email system to solicit participation from …*
		In order to maximize the available sample, (aim)	*I*		*use*	*single imputation techniques to estimate missing family income and …*
			I	*chose to*	*take a step back and examine*	*one of the untested assertions underlying the field.*
	(2). Semantic-functional pattern:	**specifier**	**research processor**	**(hinge)**	**processing action**	**thing processed + (specifier)**
	Lexico-grammatical pattern: (28 instances)	subordinating conjunction	*I*		research act verb	nominal + (aim/scope/comparator/etc.)
	Examples:	*…, because*	*I*	*need to later*	*merge*	*my data with the Fixed Income Securities Database (FISD)*
		If	*I*	*cannot*	*match*	*a subsidiary to a business segment, I attempt to match it to an …*
		…, except	*I*		*exclude*	*all utilities from the sample*
	(3). Semantic-functional pattern:	**(specifier)**	**research processor**	**(hinge) + processing action**		**thing processed + (specifier)**
	Lexico-grammatical pattern: (36 instances)	(adverbial)	*I*	cognition act verb		nominal + (aim/manner)
	Examples:	*To test this empirical hypothesis,* (aim)	*I*	*estimate*		*M-Models 4 and 8, which are identical to M-Models 1 and 5…*
			I	*also consider*		*the correlates of partisan strength…*
		In treating all debt levels as equivalent, (manner)	*I*	*ignore*		*a large amount of information.*

Table 7
Local grammar analyses of I as opinion-holder/originator.

Function 4: ***I as opinion-holder/ originator*** **(227 instances)**	**(1).** Semantic-functional pattern:	**(specifier)**	**opinion-holder/ originator**	**(hinge)**	**opinion-holding/ originating action**	**opinion /thing to hold opinion on**
	Lexico-grammatical pattern: (172 instances)	(linking adverbial)	*I*	(modal verb *would*/ verb of intention)	stance-related verb/adj./noun	object clause / nominal + (complement/ appositive clause)
	Examples:	*For this reason,* (result/ inference)	*I*		*assume*	*that the published articles meet native-speaker standards for their respective languages.*
		However, (contrast)	*I*	*would*	*argue*	*that what is currently needed is a critical theory of place naming that takes into account …*
			I		*agree with*	*this distinction.*
	(2). Semantic-functional pattern:	**opinion,**	**opinion-holder/ originator**	**(hinge) + opinion-holding/originating action,**		**opinion**
	Lexico-grammatical pattern: (34 instances)	main clause	*I*	stance verb		main clause
	Examples:	*And here,*	*I*	*am claiming,*		*reliability could mean something much different than statistical…*
		It is most natural,	*I*	*would suggest,*		*to view the situation as a sort of symbiosis.*
		This examination of some of the archival evidence shows,	*I*	*believe,*		*that there is some intriguing evidence for the supposition that Moore's views at this period were…*
	(3). Semantic-functional pattern:	**specifier - topic**	**(hinge)**	**opinion-holder/ originator**	**(hinge) + opinion-holding/ originating action**	**(opinion) + (specifier)**
	Lexico-grammatical pattern: (16 instances)	nominal	(relative pronoun)	*I*	discourse act verb/ delexicalized verb	(nominal) + (adverbial)
	Examples:	*The point*		*I*	*want to make*	*here …*
		This enterprise,	*which*	*I*	*term*	*'crustal mobility',*
		…better decisions,	*which*	*I*	*interpret*	*as better inflationary outcomes,*

John (1999), this role is the most powerful role, as the author exerts ownership over his or her writing by "claiming authority as the source of the content of the text and showing that they perceive themselves as people who have the right and the ability to hold an opinion or to originate a new idea" (ibid: 29). Although not as frequent as the role of *I as recounter of research process*, this role is still

highly common, suggesting that the soft science writers do not shy away from making overt authorial presence to take a stand or put forward a view. The role of *I as architect* ranks third, with 131 instances (18.82%). This usage of the authorial *I* foregrounds the person who organizes, structures, and outlines the content of an article. This role shows the least powerful authorial presence. *I as data analyser* is the smallest functional type of *I*, with only 36 instances (5.17%). This role foregrounds the person who finds out factual result based on analysis of data, which can then be used as a basis for agreeing, disagreeing, or putting forward a claim, view, etc. Compared with other roles, soft science writers are less likely to enact this role by using *I*, but tend more to use less direct ways, such as the use of dummy-*it* constructions (157 instances), to state factual information in data analysis.

Tables 4–7 also provide an overview of the local grammars of the authorial *I* for each of the four functions, mapping semantic-functional patterns on to lexico-grammatical patterns. It should be noted that only the local grammar patterns with more than 3 occurrences of *I*-sentences are displayed in the tables. The local grammar patterns in each table represent the most frequent and salient patterns, but not necessarily an exhaustive representation of all the patterns that are found to realise each function. More specifically, overall, the three local grammar patterns in Table 4 can cover 98.47% ((96 + 17+16)/131) of the total occurrences of *I as architect*; the one pattern displayed in Table 5 represents 97.22% (35/36) of the occurrences of *I as data analyser*; the three patterns in Table 6 represent 98.68% ((234 + 28+36)/302) of the occurrences of *I as recounter of research process*; and the three patterns in Table 7 represent 97.80% ((172 + 34+16)/227) of the cases of *I as opinion-holder/originator*.

(1) Function 1: *I as architect*

When enacting the role of *I as architect*, the authorial *I* is found to typically co-occur with the "locator" in three main semantic-functional patterns: (1) "locator, architect + (hinge) + architecting action + thing architected + (aim)", (2) "hinge + architect + (hinge) + architecting action + locator, thing architected", and (3) "thing architected + (hinge) + architect + architecting action + locator" (Table 4).

Semantic-functional pattern (1), with 96 instances, is the one most frequently used to realise this function. It accounts for 73.28% (96/131) of the occurrences of *I* to enact the role of *I as architect*. In the pattern, *I* occurs in the main clause, as in the example sentence *In this section, I examine the extent to which subsidiary debt protects a segment's cash flows from getting diverted* …. The "locator" consists of three types of adverbials: text deixis (e.g. *in* section 5), listing adverbial (e.g. *firstly*), and time adverbial (e.g. *now*). The "hinge" consists of the modal verbs (e.g. *will*, *shall*, *have*) and the verbs of intention (e.g. *attempt to*, *seek to*, *wish to*). The "architecting action" is commonly realised by research act verbs (e.g. *use, examine, explore, show, demonstrate, undertake, provide*) and discourse act verbs (e.g. *elaborate, describe, discuss, present, suggest, review, begin, conclude*), and occasionally by the cognition act verbs *consider* and *focus on*. The "thing architected" is mainly realised by noun phrases or *wh*-clauses (the subordinate clause that's introduced by one of the *wh*-words). When realised by noun phrases, the "thing architected" is often followed by *to*-infinitive clause to indicate the aim of research (e.g. *In the next section, I explore the potential of the concept of habitus to explain the difficulty changing folk pedagogies.*). In line with the description above, the lexico-grammatical pattern for semantic-functional pattern (1) is generalized as "text deixis/time/listing adverbial + *I* + (modal verb/verb of intention) + verb of research/discourse/cognition act + [nominal + (to-inf)]/*wh*-clause".

Semantic-functional pattern (2) occurs 17 times in total, accounting for 12.98% of the occurrences of *I as architect*. In this pattern, the authorial *I* occurs in the subordinate clause introduced by *as*, as in the example sentence *As I argue below, laws may express disgust: anti-sodomy laws in the United States*. The "locator" consists typically of two types: text deixis (e.g. *below*, *above*) and time adverbial (e.g. *earlier*). The "architecting action" is mainly realised by discourse act verbs (e.g. *note, argue, describe, discuss, suggest*), and in a few cases by the research act verb *demonstrate*. Thus, semantic-functional pattern (2) can be matched with the lexico-grammatical pattern "*as* + *I* + (modal verb) + verb of discourse/research act + text deixis/time adverbial + main clause".

Semantic-functional pattern (3), with 16 instances (12.21%), is a common variant of pattern (2). In the pattern, *I* occurs in the attributive clause of the "thing architected", as in the example sentence *The results I present here demonstrate that federalism does play a part in*. The "locator" consists mostly of text deixis (e.g. *here*, *above*), and the "architecting action" consists mainly of discourse act verbs (e.g. *present, describe, discuss*), forming the lexico-grammatical pattern "nominal + attributive clause{(that) + *I* + verb of discourse act + text deixis adverbial} + main clause".

(2) Function 2: *I as data analyser*

When enacting the role of *I as data analyser*, the authorial *I* shows a strong tendency to co-occur with the "specifier". 35 out of 36 instances of *I* in this role (97.22%) are found to occur in the semantic-functional pattern: "specifier + data-analyser + (negation) + analysing action + thing found + (specifier)" (Table 5). Six main types of specifiers are found in the pattern: scope (e.g. *in untabulated results*), manner (e.g. *using public filings*), result/inference (e.g. *hence*), addition (e.g. *furthermore*), listing (e.g. *finally*), and contrast (e.g. *however*). The "negation hinge" consists of the negative verb *fail* and adverb *not* to indicate the unexpected or unsatisfactory result. The "analysing action" consists of verbs of perception (e.g. *observe*, *see*, *note*, *find*) and verbs of verification (e.g. *confirm*, *verify*). The "thing found" consists of the object clause, the noun phrase, and the noun with appositive clause (e.g. *evidence that* …, *fact that* …). In accordance with the analysis, the lexico-grammatical pattern is generalized as "(source/manner/result/inference/addition/listing/contrast) adverbial + *I* + (negative verb) + verb of perception/verification + [object clause/noun + (appositive clause)]".

(3) Function 3: *I as recounter of research process*

When enacting the role of *I as recounter of research process*, the authorial *I* tends to occur in three main semantic-functional patterns:

(1) "(specifier) + research processor + (initiative) + processing action + thing processed + (specifier)", (2) "specifier + research processor + (hinge) + processing action + thing processed + (specifier)", and (3) "(specifier) + research processor + (hinge) + processing action + thing processed + (specifier)" (Table 6).

Semantic-functional pattern (1) is the most common one to realise the function. Its frequency reaches 234 times, amounting to 77.48% of the occurrences of *I* in this role. The pattern is signalled with the predominant use of research act verbs (e.g. *ask, choose, compute, conduct, demonstrate, develop, distinguish, divide, examine, exclude, explore, extend, generate, give, identify, hand-check, include, interview, investigate, keep, label, leave, limit, match, obtain, provide, run, rerun, separate, take, use*). The "initiative hinge" (e.g. *choose to, decide to, attempt to, want to*) is typically found to co-occur with the research act verb to indicate how committed the author is to his/her methodological decisions. Various types of the "specifier" are shown in the pattern, such as aim (e.g. *In order to maximize the available sample*), manner (e.g. *By using mean ROA instead of realised ROA*), scope (e.g. *for each industry in each year*), result/inference (e.g. *therefore*), addition (e.g. *Barring that*), listing (e.g. *next*), emphasis (e.g. *in particular*) and contrast (e.g. *however*). The "thing processed" is commonly realised by the noun phrase, often followed by the "specifier", such as aim (e.g. *I use multinomial logistic regression analysis to model the probability of …*), method (e.g. *I therefore compute all standard errors using the cluster method*), scope (e.g. *I test these predictions on my sample of nonguaranteed subsidiary*), and comparator (e.g. *I match subsidiaries with debt outstanding to divisions of firms*). The lexico-grammatical pattern is thus summarized as "(addition/aim/contrast/emphasis/manner/listing/scope/ result/inference adverbial) + *I* + (verb of decision/intention) + research act verb + nominal + (aim/ method/scope/comparator/etc.)".

Semantic-functional pattern (2) occurs 28 times (9.27%). In the pattern, *I* occurs in the adverbial clause, in particular, introduced by the subordinating conjunction *because, except, if, when,* and *as*, as in the example sentence … *except I exclude all utilities from the sample*. Like semantic-functional pattern (1), the "processing action" of this pattern consists mostly of the research act verbs as exemplified above, and the "thing processed" is mainly realised by the noun phrase, typically followed by the "specifier" of aim, scope, and comparator. However, different from semantic-functional pattern (1), verbs of decision (e.g. *choose to, decide to*) are not found in this pattern, and verbs of intention (e.g. *wish to, want to, need to*) are only found in the adverbial clause of reason (e.g. *because I need to later merge my data with the Fixed Income Securities Database (FISD)* …) but not in other types of adverbial clause. The lexico-grammatical pattern for semantic-functional pattern (2) is generalized as "subordinating conjunction + *I* + research act verb + nominal + (aim/ scope/comparator/etc.)".

Semantic-functional pattern (3), with 36 instances (11.92%), features the use of cognition act verbs (e.g. *consider, estimate, ignore, focus on*) to perform the "processing action", as in the example sentence *In treating all debt levels as equivalent, I ignore a large amount of information*. The "specifier" in this pattern consists of addition, aim, contrast, emphasis, listing, manner, etc., and the "thing processed" consists mostly of noun phrases, occasionally followed by the aim or manner "specifier" (e.g. … *I estimate them by subtracting capital expenditures from end-of-period assets and adding depreciation.*). Semantic-functional pattern (3) can be matched with the lexico-grammatical pattern "(addition/aim/contrast/emphasis/listing/manner adverbial) + *I* + cognition act verb + nominal + (aim/ manner)".

(4) Function 4: *I as opinion-holder/originator*

The identity role of *I as opinion-holder/originator* is most often realised in three semantic-functional patterns: (1) "(specifier) + opinion-holder/originator + (hinge) +opinion-holding/originating action + opinion", (2) "opinion, opinion-holder/originator + (hinge) + opinion-holding/originating action, opinion", and (3) "topic/opinion + (hinge) + opinion-holder/originator + (hinge) + opinion-holding/originating action + (opinion) + (specifier)" (Table 7).

Semantic-functional pattern (1), with 172 occurrences (75.77%), is the one most frequently used to express an opinion. The "specifier" in the pattern consists of various types of adverbials, in particular, result/inference (e.g. *for this reason*), addition (e.g. *on the other hand*), aim (e.g. *To simplify matters*), contrast (e.g. *by contrast*), manner (e.g. *in this way*), scope (e.g. *In the case of conflict avoidance*), and text deixis (e.g. *here*). The "hinge" consists largely of the "mitigator", which is realised by the modal verb *would* and the verbs of wish (e.g. *wish to, hope to*), and the "initiative", which is realised by the verbs of intention (e.g. *want to, prefer to, propose to*). The "opinion-holding/originating action" consists mainly of stance-related words or constructions, which include cognition act verbs (e.g. *assume, believe, conjecture, expect, fear, hypothesize, predict, presume, think, contend, find, agree, expect, consider, prefer*), discourse act verbs (e.g. *posit, propose, suggest, define, postulate, state*), delexicalized verbs collocating with stance nouns (e.g. *make no such assumption, did not have any a priori predictions, offer the following proposal*), and predicate adjectives (e.g. *be equally/reasonably confident that, remain true to*). The "opinion" is realised in two main structures: the object clause (e.g. *I argue that historical contingency had an impact on how models were constructed* …) and the noun phrase that is sometimes followed by the complement (e.g. *I consider some of them to be like more than just a business relationship* …) or the appositive clause (e.g. *I have no doubt, indeed we have seen it here, that other approaches to recognition can provide accurate accounts of some of the findings that we explored.*). In line with the analysis, the lexico-grammatical pattern is generalized as "(result/inference/addition/ aim/contrast/manner/scope/text deixis adverbial) + *I* + (*would*/verb of intention) + stance-related verb/ adjective/noun + [object clause/nominal + (complement/appositive clause)]".

Semantic-functional pattern (2) occurs 34 times (14.98%) and ranks second in frequency. The pattern functions as the parenthesis that is inserted into a main clause, as in the example sentence *One reason, I suggest, is that the psychiatrist is the contemporary arbiter of abnormality*. The "opinion-holding/originating action" is mostly realised by stance-related verbs, but compared with those occurring in semantic-functional pattern (1), the variety of verbs in this pattern is much limited, consisting mostly of *believe, claim, hope, maintain, think,* and *suggest*. The "mitigator" *would* is occasionally found to precede the stance-related verb, while the "initiative" is not shown in this pattern. The lexico-grammatical pattern for semantic-functional pattern (2) is thus "main clause, parenthesis{*I* + (*would*) + stance verb}, main clause".

Semantic-functional pattern (3), with 16 instances (7.05%), is the least common one, but not to be missed, for this function. It functions as the attributive clause with the "topic specifier" as its head noun, as in the example sentence *This enterprise, which I term 'crustal mobility', was concerned with* …. The most frequent verbs to perform the "opinion-holding/originating action" in the pattern are discourse act verbs (e.g. *call*, *term*, *interpret*) and the delexicalized verb *make*. Only the "initiative hinge" *want to* is found in this pattern, and it only occurs with the head noun *point*, as in the sentence *The point I want to make here is that* …. The lexico-grammatical pattern can be summarized as "head noun + attributive clause{(*that/which*) + *I* + discourse act verb/delexicalized verb + (nominal) + (adverbial)} + main clause".

4.2. Co-selection of functions and patterns of *I*

The local grammar analysis reflects the mappings or co-selections of functions, meanings, and patterns. To be more specific, the authorial *I* tends to collocate with certain verbs and to occur in certain lexico-grammatical patterns and semantic-functional patterns in order to realise particular functions, and a specific function also promotes and restricts the use of *I* in certain patterns and with certain collocations. Co-selection underlying the use of *I* prevails into multiple levels and can be elaborated on from three aspects.

(1) Co-selection of lexis and lexis. The Sinclairian idiom principle accounts for the phraseological tendency, viewing language as essentially being made up of strings of co-selected words; in other words , the selection of one word restricts the selection of another. Our analyses accord with this view. For example, as shown in Table 4 above, when enacting the role of *I as architect*, the authorial *I* in semantic-functional pattern (2) occurs mostly with discourse act verbs, occasionally with research act verbs, and hardly with cognition act verbs. For another example, despite the large number of stance-related verbs, the verbs that occur with the *opinion-holder/originator I* in semantic-functional pattern (2) are highly limited, consisting mostly of *believe, claim, hope, maintain, think,* and *suggest* (Table 7). Although the co-occurrences of *I* and its following verb are not fully formulaic, they represent preferred ways of achieving particular functions in academic writing. These tendencies for preferred combinatory behaviour of the authorial *I* and its following verb reflect the co-selection between lexis and lexis: once *I* is selected for a given function, its following verb is simultaneously constraint to a varying extent.

(2) Co-selection of lexis and grammar. Co-selection points to the integrity of lexis and grammar (Sinclair, 2006): a word has its preferred grammatical structure for making meaning, and a grammatical structure selects a limited range of lexical words for achieving a function. For example, "(addition/aim/contrast/emphasis/manner/listing/ scope/result/inference adverbial) + *I* + (verb of decision/intention) + research act verb + nominal + (aim/method/scope/comparator/etc.)" is the most used to realise the function of recounting research process (Table 6). The pattern consists of lexical words (e.g. *I*), grammatical categories (e.g. adverbial, verb), semantic categories (e.g. research act verb), and functional categories (e.g. aim/method/scope/comparator), so they are not purely grammatical or lexical. Patterns of this kind are the product of co-selection of lexis and grammar. It is argued that words in creating meaning naturally attract other words to their company, and the linear arrangement of words inevitably result in phraseologies that incorporate lexis and grammar. In other words, the linear arrangement of language entails the co-selection of lexis and lexis and the co-selection of lexis and grammar on the syntagmatic axis (Sinclair, 2006).

(3) Correspondence of formal elements and meaning elements in a local grammar. By identifying the semantic-functional pattern and the lexico-grammatical pattern of each *I*-sentence, local grammar analyses help to establish the connection between meaning elements and formal elements. For example, in the analysis of *I as architect*, three main semantic-functional patterns have been identified: (1) "locator, architect + (hinge) + architecting action + thing architected + (aim)", (2) "hinge + architect + (hinge) + architecting action + locator, thing architected", and (3) "thing architected + (hinge) + architect + architecting action + locator". The three semantic-functional patterns correspond to the three lexico-grammatical patterns, respectively: "text deixis/time/listing adverbial + *I* + (modal verb/verb of intention) + verb of research/discourse/cognition act + [nominal + (to-inf)]/*wh*-clause", (2) "*as* + *I* + (modal verb) + verb of discourse/research act + text deixis/time adverbial + main clause", and (3) "nominal + attributive clause{(that) + *I* + verb of discourse act + text deixis adverbial} + main clause" (Table 4). The high degree of correspondence between lexico-grammatical patterns and semantic-functional patterns suggests that the co-selection between meaning elements and formal elements is highly possible for *I*-sentences to achieve a function in soft science articles. The mappings of formal elements and meaning elements in local grammar analyses also demonstrate that the relationship between form and meaning is far from arbitrary. Each word occurs in a limited set of patterns, and words which share pattern also share meaning (Hunston & Sinclair, 2000). In other words, particular forms support or modify pertinent meanings, and particular meanings correlate with and select certain forms. Conceived of in this way, it is argued that not the single word *I* but *I* and its local grammar patterns (in terms of both lexico-grammatical patterns and semantic-functional patterns) are central in meaning creation. The exercise of local grammar analyses enables us to view text production as "not just the strings of co-occurring phraseological forms but also strings of discourse-semantic patterns" (Su & Zhang, 2019, p. 15), in so doing, local grammars help to establish a more transparent connection between form and meaning.

5. Conclusion

Based on the analysis of 696 authentic occurring instances of the authorial *I* in soft science articles, this paper has examined the links between functions and patterns of *I* from the local grammar approach. The functions of *I*, in terms of the writer role enacted by *I*, are divided into four groups: *I as architect*, *I as data analyser*, *I as recounter of research process*, and *I as opinion holder/originator*. In the

local grammar analyses of the four functions of *I*, we have generalized the semantic-functional patterns of *I* to realise each function, the frequent lexico-grammatical patterns for each semantic-functional pattern, and the collocating verbs in each lexico-grammatical pattern.

This study has extended the local grammar analyses from starting from a function to starting from a restricted form, by making the case that local grammars can be explored to more fully describe the combinatory behaviour of the authorial *I*. It has been illustrated that the local grammar approach is better at describing expressions of meaning, as it captures "both the lexical-grammatical resources and their associated discourse-semantic patterns, thereby offering an inventory of linguistic strategies available to language users to express specific meanings in communication" (Su & Zhang, 2019, p. 1). The co-selection among patterns, meanings, and functions of *I* is going on at multiple levels: the co-selection of lexis and lexis, the co-selection of lexis and grammar, and the co-selection of patterns and functions. Those co-selective relationships have provided strong evidence that form and meaning are inseparable from each other, and linguistic description should keep to the unity of form and meaning. In Sinclair's words, "there is ultimately no distinction between form and meaning" (Sinclair, 1991, p. 7).

The study has also found that, as a sign of strong authorial presence, the authorial *I* is not absent from soft science writing. In contrast, the occurrences of *I* are unignorable. However, the issue is not as simple as how frequently soft science writers should use *I*. As noted by Lee and Swales (2006: 57), "what apprentice writers may be mostly missing is fine tuning of lexical and syntactic subtleties, particularly in terms of their strategic and rhetorical implications". A knowledge of the strategic use of *I* is thus of great value to apprentice writers, as "they must know, in the process of writing an article for publication, how to emphasize their personal contributions to their field of research and how to seek cooperation and stress solidarity with expected readers and their disciplines" (Kuo, 1999, p. 136). Scrutiny of the local grammars of the authorial *I* captures more of the subtleties of the co-selection patterns of *I* and provides detailed and contextualized insights into the lexico-grammatical and discourse-semantic constraints on the strategic use of *I*. It is hoped that a better understanding of the co-selection between functions and patterns of the authorial *I* based on this study would help soft science novice writers to construct their authorial identity more strategically and effectively.

This article has identified the local grammars of the authorial *I* in soft sciences in general, following the hard-soft distinction that "offer[s] a useful way of examining general similarities and differences between fields" (Hyland, 2004, p. 93). One factor which has not yet been considered in the analysis, but which may exert an influence on some of the results obtained here, is the possible inter- or intra-disciplinary variations in the use of *I* within soft sciences. Some of the findings may not be able to bring discipline specificity to self-mention resources, but this article shows the diversity of approaches (e.g. local grammar in this study) to exploring the co-selection patterns of the authorial *I* and contribute methodologically to the establishment of connections among form, meaning and function. Further research can investigate the local grammars of *I* in different soft science disciplines or in different types of research within a discipline (e.g. qualitative research versus quantitative research). In addition, the corpus used in this study is a comparable corpus of English research articles written by English and Chinese native speakers. Following this study, we also intend to explore the differences and similarities in the use of *I* by L1 English and L1 Chinese writers in order to better understand the complex interplay between national cultures and disciplinary cultures (Molino, 2010) regarding the use of *I*.

CRediT authorship contribution statement

Jingjie Li: Conceptualization, Methodology, Investigation, Formal analysis, Validation, Writing – original draft, Writing – review & editing. **Ting Ye:** Investigation, Formal analysis, Validation, Writing – original draft, Writing – review & editing.

References

Allen, C. M. (2005). *A local grammar of cause and effect: A corpus-driven study*. Unpublished PhD dissertation. University of Birmingham.

Barnbrook, G. (2002). *Defining language: A local grammar of definition sentences*. Philadelphia: John Benjamins Publishing Company.

Barnbrook, G., & Sinclair, J. (2001). Specialized corpus, local and functional grammars. In M. Ghadessy, A. Henry, & R. L. Roseberry (Eds.), *Small corpus Studies and ELT. Theory and practice, 237–276*. Amsterdam: John Benjamins.

Bednarek, M. (2008). *Emotion talk across corpora*. London: Palgrave.

Cheng, W., & Ching, T. (2018). 'Not a guarantee of future performance': The local grammar of disclaimers. *Applied Linguistics, 39*(3), 263–301.

Dong, T., & Qiu, L. (2018). Research on self-mentions and author identity in academic English writing. *International Journal of English Language and Literature Studies, 7*(4), 115–121.

Dueñas, P. M. (2007). 'I/we focus on…': A cross-cultural analysis of self-mentions in business management research articles. *Journal of English for Academic Purposes, 6*(2), 143–162.

Fortanet, I. (2004). The use of *we* in university lectures: Reference and function. *English for Specific Purposes, 23*(1), 45–66.

Gross, M. (1993). Local grammars and their representation by finite automata. In M. Hoey (Ed.), *Data, description, discourse. Papers on the English language in honour of John McH. Sinclair* (pp. 26–38). London, UK: Harper-Collins.

Halliday, M. A. K. (1994). *An introduction to functional grammar* (2nd ed.). London: Edward Arnold.

Harwood, N. (2005a). 'I hoped to counteract the memory problem, but I made no impact whatsoever': Discussing methods in computing science using I. *English for Specific Purposes, 24*(3), 243–267.

Harwood, N. (2005b). 'Nowhere has anyone attempted… In this article I aim to do just that' A corpus-based study of self-promotional *I* and *We* in academic writing across four disciplines. *Journal of Pragmatics, 37*(8), 1207–1231.

Harwood, N. (2005). We do not seem to have a theory… The theory I present here attempts to fill this gap': Inclusive and exclusive pronouns in academic writing. *Applied linguistics, 26*(3), 343–375.

Hunston, S. (2002). Pattern grammar, language teaching, and linguistic variation: Applications of a corpus-driven grammar. In R. Reppen, S. Fitzmaurice, & D. Biber (Eds.), *Using Corpora to explore linguistic variation, 167–183*. Amsterdam: John Benjamins.

Hunston, S., & Sinclair, J. (2000). A local grammar of evaluation. In S. Hunston, & G. Thompson (Eds.), *Evaluation in text: Authorial stance and the construction of discourse* (pp. 74–101). New York: Oxford University Press.

Hunston, S., & Su, H. (2017). Patterns, constructions, and local grammar: A case study of 'evaluation'. *Applied Linguistics*, 1–28.

Hyland, K. (2002). Authority and invisibility: Authorial identity in academic writing. *Journal of Pragmatics, 34*(8), 1091–1112.

Hyland, K. (2004). A convincing argument: Corpus analysis and academic persuasion. In U. Connor, & T. A. Upton (Eds.), *Discourse in the professions: Perspectives from corpus linguistics* (pp. 87–112). Amsterdam: John Benjamins.
Hyland, K., & Jiang, F. (2017). Is academic writing becoming more informal? *English for Specific Purposes, 45*, 40–51.
Hyland, K., & Tse, P. (2005). Hooking the reader: A corpus study of evaluative that in abstracts. *English for Specific Purposes, 24*(2), 123–139.
Isik-Tas, E. (2018). Authorial identity in Turkish language and English language research articles in Sociology: The role of publication context in academic writers' discourse choices. *English for Specific Purposes, 49*, 26–38.
Ivanic, R. (1998). *Writing and identity: The discoursal construction of identity in academic writing*. Amsterdam: John Benjamins Publishing Company.
Kim, C. (2009). Personal pronouns in English and Korean texts: A corpus-based study in terms of textual interaction [J]. *Journal of Pragmatics, 41*, 2086–2099.
Kolb, D. A. (1981). Learning styles and disciplinary differences. In A. Chickering (Ed.), *The modern American college* (pp. 232–255). San Francisco: Jossey Bass.
Kuo, C. H. (1999). The use of person pronouns: Role relationships in scientific journal articles. *English for Specific Purposes, 18*(2), 121–138.
Leedham, M., & Fernandez-Parra, M. (2017). Recounting and reflecting: The use of first-person pronouns in Chinese, Greek and British students' assignments in engineering. *Journal of English for Academic Purposes, 26*, 66–77.
Lee, D., & Swales, J. (2006). A corpus-based EAP course for NNS doctoral students: Moving from available specialized corpora to self-compiled corpora. *English for Specific Purposes, 25*(1), 56–75.
McGrath, L. (2016). Self-mentions in anthropology and history research articles: Variation between and within disciplines. *Journal of English for Academic Purposes, 21*, 86–98.
Molino, A. (2010). Personal and impersonal authorial references: A contrastive study of English and Italian linguistics research articles. *Journal of English for Academic Purposes, 9*(2), 86–101.
Raušová, V. (2018). Pragmatic functions of I in academic discourse: Linguistic research articles. *Linguistica Pragensia, 28*(2), 168–183.
Sinclair, J. (1991). *Corpus, concordance and collocation*. Oxford: Oxford University Press.
Sinclair, J. (2006). *Linear unit grammar: Integrating speech and writing*. Amsterdam/Philadelphia: John Benjamins Publishing Company.
Sinclair, J. (2007). Defining the definiendom. In G. de Schryver (Ed.), *A way with words: Recent advances in lexical theory and analysis: A Festschrift for Patrick Hanks* (pp. 37–47). Kampala, Uganda: Menha Publishers.
Smith, T. R. (1996). *Writing in psychology* (2nd ed.). New York: Wiley [M].
Spencer, C., & Arbon, B. (1996). *Foundations of writing: Developing research and academic writing skills [M]*. Lincolnwood, IL: National Textbook Company.
Starfield, S., & Ravelli, L. J. (2006). "The writing of this thesis was a process that I could not explore with the positivistic detachment of the classical sociologist": Self and structure in New Humanities research theses. *Journal of English for Academic Purposes, 5*(3), 222–243.
Su, H. (2017). Local grammars of speech acts: An exploratory study. *Journal of Pragmatics, 111*, 72–83.
Su, H. (2018). Is a meaning-based grammar of English possible? Exploring English grammar from the perspectives of pattern grammar and local grammar. *English Today, 35*(4), 1–7.
Su, H. (2019). Patterns, local grammars, and the design of English teaching materials. *ELT Journal, 74*(1), 1–10.
Su, H., & Wei, N. (2018). "I'm really sorry about what I said": A local grammar of apology. *Pragmatics, 28*(3), 439–462.
Su, H., & Zhang, L. (2019). Local grammars and discourse acts in academic writing: A case study of exemplification in linguistics research articles. *Journal of English for Academic Purposes, 43*, 1–25.
Tang, R., & John, S. (1999). The 'I' in identity: Exploring writer identity in student academic writing through the first person pronoun. *English for Specific Purposes, 18*, 23–39.
Taylor, H., & Goodall, J. (2019). A preliminary investigation into the rhetorical function of 'I'in different genres of successful business student academic writing. *Journal of English for Academic Purposes, 38*, 135–145.
Vassileva, I. (1998). Who am I/who are we in academic writing: A contrastive analysis of authorial presence in English, German, French, Russian and Bulgarian. *International Journal of Applied Linguistics, 8*(2), 162–190.
Wang, M., & Zhang, Y. (2021). 'According to…': The impact of language background and writing expertise on textual priming patterns of multi-word sequences in academic writing. *English for Specific Purposes, 61*, 47–59.
Warren, M., & Leung, M. (2016). Do collocational frameworks have local grammars? *International Journal of Corpus Linguistics, 21*(1), 1–27.
Wei, N., & Zhang, L. (2020). Introducing Beijing CARE academic English corpus. *Corpus Linguistics, 7*, 72–77, 01.
Yeo, J., & Ting, S. (2014). Personal pronouns for student engagement in arts and science lecture introductions. *English for Specific Purposes, 34*(1), 26–37.
Zareva, A. (2013). Self-mention and the projection of multiple identity roles in TESOL graduate student presentations: The influence of the written academic genres. *English for Specific Purposes, 32*(2), 72–83.

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